

Clinical-Bladder cancer

What motivates bladder cancer patients to be active? A qualitative study assessing attitudes towards physical activity and digital health technologies

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Abstract

Introduction and Objective: Patients undergoing neoadjuvant chemotherapy (NAC) and radical cystectomy (RC) are often older, frail and deconditioned, leaving them with limited reserve to endure the demands of treatment. As a result, improving physical fitness in this population remains a critical need. Two emerging means to do so include the implementation of exercise programs pre- and postsurgery (i.e. (p) rehabilitation), and the use of digital health technologies (i.e. Fitbit®, Apple Watch™, exercise apps) to incentivize physical activity. However, little is known about the motivations of NAC/RC patients to be active or their perceptions towards digital health technologies.

Methods: We conducted semi-structured interviews with bladder cancer survivors who previously underwent NAC/RC. We asked participants to reflect upon their experience throughout the course of treatment as it pertained to (1) barriers of physical activity, (2) facilitators of physical activity, and (3) how digital health technologies can promote physical activity. These findings will inform the design of a Phase II exercise prehabilitation trial (GetMoving, NCT06040762).

Results: 9 patients were interviewed and common themes emerged. Patients undergoing NAC/RC desired a structured exercise plan outlining the duration, intensity and frequency of exercise to perform. Encouragement to exercise and “check ins” from their provider were critical in maintaining their motivation to exercise. Objective measurements of their fitness would help with setting fitness goals and adhering to training plans. All patients would use digital health technologies to improve their health.

Conclusions: We provide a novel study assessing the perceptions of bladder cancer survivors towards exercise motivations and digital health technologies. Published by Elsevier Inc.

Key Words: Physical activity; Digital health technology; Exercise; Neoadjuvant chemotherapy; Radical cystectomy; Muscle invasive bladder cancer

1. Introduction

Patients with bladder cancer undergoing neoadjuvant chemotherapy (NAC) and radical cystectomy (RC) experience significant morbidity. These patients are often older with a high burden of frailty and deconditioning, leaving them with limited physiologic reserve to endure the

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demands of treatment [1]. As a result, improving physical fitness and functional reserve in those undergoing NAC/RC has emerged as a central objective in optimizing the urological care of this population [2,3]. Prehabilitation and rehabilitation programs have shown promise in addressing this need by improving fitness and candidacy for invasive therapies while accelerating recovery after cystectomy [3–5]. Additionally, digital health technologies such as wearable devices (e.g. FitBit® or Apple™ Watch) and exercise apps have demonstrated success in enacting lifestyle changes, increasing step count, facilitating physical activity participation, and providing psychosocial support to exercise [6,7].

Despite these developments, little is known about the perspectives of patients undergoing NAC/RC regarding exercise motivations and digital health technologies. Few studies have explored the factors and motivations that influence their decisions to engage in or avoid physical activity during oncologic treatment. Additionally, research on patient-reported outcomes (PROs) for digital health technologies has primarily assessed how these tools can track complications and prevent hospital readmissions, rather than how these technologies can support activity adherence during and after treatment [8,9].

Given this context and in preparation for a randomized controlled trial evaluating a personalized perioperative physical activity intervention (NCT06040762), we conducted a qualitative study exploring the perspectives of bladder cancer survivors who had previously undergone NAC and RC regarding physical activity. Our aim was to identify perceived barriers and facilitators to exercise, as well as attitudes toward digital health technologies designed to enhance fitness. Understanding these viewpoints is essential for developing effective strategies to increase physical activity among NAC/RC patients and optimize the use of technology in supporting their health.

2. Materials and methods

2.1. Study design and population

Between June 2023 to July 2023, we conducted semi-structured interviews with bladder cancer survivors who previously underwent NAC and RC. Participants provided informed consent and received a \$50 gift card following interview completion. The study was approved by our Institutional Review Board (00017722). Notably, this project was undertaken to inform the design of the Phase II GetMoving Clinical Trial (NCT06040762), a randomized controlled trial comparing perioperative home-based exercises to standard care in patients undergoing RC or nephroureterectomy for localized bladder or upper tract urothelial carcinoma with or without NAC.

2.2. Quantitative measures

Demographic characteristics of participants were extracted from the electronic health record. Descriptive statistics were performed on demographic data.

2.3. Qualitative interviews

Semi-structured interviews were conducted using a video-conferencing software platform (Zoom®) by 2 study members with experience facilitating semi-structured interviews. Study members had had no prior relationship with the participants. Interviews were completed within 45 to 60 minutes and were recorded and transcribed.

Interviews were conducted using a think-aloud approach, which involves a participant verbalizing their lived experiences and perceptions. Participants were asked open-ended questions about their perceived barriers and facilitators of physical activity, as well as their attitudes towards digital health technologies. Specifically, they were asked about their perceptions and prior utilization of wearable physical activity monitors that track health metrics such as step count, heart rate, and calories burned. Common examples of these are the FitBit® or Apple™ Watch. Participants were then given the opportunity to view and provide feedback on a physical activity app called ExerciseRx. This app is in beta-testing phase and was developed by a team at University of Washington.

2.4. Qualitative analysis

Interview transcriptions were coded using *Dedoose* software (Version 7.0.23, Los Angeles, CA) [10]. Two trained qualitative analysts independently analyzed the transcripts using a process of reflexive thematic analysis, which is a well-described method of analyzing qualitative data [11]. This type of analysis relies on a researcher's subjective interpretation of patterns across a dataset, requiring researchers to engage with subjectivity and reflexivity in the research process [12]. In our analysis, we drew on our experiences as designers, health informatics researchers, and clinical researchers to interpret data and develop themes, which required us to consider that patient experiences and attitudes differed from our expectations.

3. Results

Nine participants completed an interview. Median age was 72 years and 7 participants were men (Table 1). Most were Caucasian ($n = 8$), married ($n = 7$) and endorsed at least moderate family support at home. The cohort reflected the typical population selected for NAC/RC, characterized by a high comorbidity burden and good functional status. Six participants had a Charlson Comorbidity Index score of ≥ 5 , and all had an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1.

Surgical approaches were nearly evenly split, with 4 patients undergoing open cystectomy and 5 undergoing robotic procedures. The majority ($n = 7$) received an ileal conduit for urinary diversion. Within 90 days postsurgery, nearly all patients ($n = 7$) experienced complications, including 6 who developed grade 3 treatment-related

Table 1
Patient demographics and surgical characteristics.

	No (%)
Median age in years (IQR)	72 (65–79)
Male	7 (78)
Self-reported race	
Caucasian	8 (89)
Asian	1 (11)
Self-reported ethnicity	
Hispanic or Latino	0 (0)
Not Hispanic or Latino	9 (100)
Marital status	
Married	7 (78)
Single	1 (11)
Widowed	1 (11)
Lives with	
Spouse/Partner	6 (66)
Spouse/Partner and children	2 (22)
Alone	1 (11)
Preoperative ECOG performance status	
0	7 (78)
1	2 (22)
Preoperative Charlson Comorbidity Index	
3	2 (22)
4	1 (11)
5	5 (56)
6	1 (11)
Surgical approach	
Open	4 (44)
Robotic	5 (56)
Urinary diversion	
Ileal conduit	7 (78)
Neobladder	1 (11)
Continent cutaneous reservoir	1 (11)
Pathologic stage, at radical cystectomy ^b	
0	5 (56)
I	0 (0)
II	0 (0)
III	4 (44)
IV	0 (0)
Complications within 90-days ^a	
0	2 (22)
1	0 (0)
2	1 (11)
3	6 (67)
Clinical stage, at time of interview ^b	
No evidence of disease	7 (78)
I	0 (0)
II	0 (0)
III	1 (11)
IV	1 (11)
Median days from radical cystectomy to interview (IQR)	148 (112–184)

^a Grading according to Common Terminology Criteria for Adverse Events (CTCAE) version 5.

^b Staging according to American Joint Committee on Cancer staging for bladder cancer seventh edition.

adverse events, as classified by the Common Terminology Criteria for Adverse Events (CTCAE) v5.0.

The median time from radical cystectomy to interview was 148 days (IQR: 112–184). At the time of interview, 7 participants showed no evidence of disease, while one had

stage III and another had stage IV disease. Most participants reported engaging in more than 45 minutes of physical activity per day, with walking being the most common form of exercise. Three participants had limited exercise capacity at time of interview: one due to a stomal hernia, another experienced symptomatic anemia as a result of adjuvant pembrolizumab for stage IV disease, and a third with stage III disease who reported low energy and passed away from widespread metastatic disease 5 months following the interview.

Common themes emerged regarding patient perceptions on (1) barriers to exercise, (2) facilitators of exercise, and (3) digital health care technologies.

3.1. Barriers to exercise

3.1.1. Experiencing treatment-related debilitation

Participants frequently reported that NAC/RC significantly impacted their ability to exercise. Many described chemotherapy as particularly debilitating, citing adverse symptoms including nausea, fatigue, and weakness that made it challenging to remain active. One participant stated, “Chemo beats you up... It was hard for me to get out of bed, let alone engage in physical activity. You can’t eat, and the nausea is constant.” Another shared, “During chemo, I was sick half the week. .I didn’t feel like doing anything because I had no strength.”

In addition to the physical toll they sustained, participants mentioned that the time commitment of chemotherapy infusions limited their ability to stay active. Infusions could last up to 8 hours, leaving them drained and without time or energy for exercise. One recurring theme was a lack of preparedness for the side effects of chemotherapy, with a participant noting, “The nausea was intense, and I wasn’t fully informed about how severe the effects of chemotherapy would be.”

While cystectomy also limited the ability to exercise, it seemed like many felt better prepared for the physical exhaustion it caused, which was described as more physical than mental. One participant explained, “My mind was ready to go, but my body needed to catch up.” However, the surgery still had a profound effect on physical activity. “After surgery, I didn’t do any exercise for a couple of months. I was constantly tired, and even short walks would exhaust me.” Another added, “I wasn’t feeling good after surgery... Bending and twisting didn’t come as easily as before.”

3.1.2. Lack of exercise-related follow-up from providers

Participants reported feeling that healthcare providers did not prioritize discussions about physical activity after treatment. Post-treatment conversations typically centered on oncological outcomes, leaving exercise as a secondary concern. As 1 participant remarked, “I wouldn’t say the follow-up on physical activity was meaningful. No one asked

if I was getting up, moving around, or how fatigued I was. No one asked, ‘How often do you take a nap?’”

This lack of “check-ins” or encouragement to stay active caused many participants to deprioritize exercise. For some, the absence of physical activity guidance from their provider signaled that it was not a critical part of their recovery. As a result, they were left to find their own motivation to exercise, but as one admitted, “Sometimes there was none.”

Some participants also expressed feeling unsupported during their recovery process. One shared, “There was no one to guide me to say, ‘This is where you are in your recovery, and this is what you can do. We will support you along the way.’” Participants commented that this absence of support left them feeling “scared” or “alone,” while for others, it led to confusion about how to safely reintroduce exercise into their routine. As another participant explained, “After surgery, I didn’t know what to do. How soon could I start exercising? With this new urostomy, I wasn’t sure what exercises were safe or how to protect my incision. That’s why it took me so long to get back to the gym.”

3.2. Facilitators of exercise

3.2.1. Explicit instructions on type and duration of exercise (s) to perform

Nearly all participants expressed a desire for a *structured workout routine* that clearly outlined which exercises to perform each day, along with the specific repetitions or duration. They wanted precise guidance so that “there’s no guesswork, decisions, or choices” regarding their exercise regimen. Participants viewed exercise much like a medication prescription from their doctor—ready and willing to follow it, as long as they knew “what exercise and how much was prescribed.”

Having a well-defined workout plan “inspires patients,” providing them with something concrete to follow and aim for. Many participants in this study identified as goal-oriented individuals, and endorsed that receiving a workout plan “in black and white” would allow them to “check the box” each day, knowing they had completed their prescribed routine. Several participants emphasized that they “perform better when there’s accountability, and when clear expectations are set.” As 1 participant put it, “If there’s a plan, I’m all in.”

3.2.2. Encouragement from their doctor to exercise

Participants consistently reported that even simple reminders from their doctors to exercise had a meaningful impact on their motivation to stay active. A casual suggestion like, “You should probably move around a little more than you are,” often provided the push they needed. Positive reinforcement from physicians played a key role in encouraging healthy exercise habits. For example, comments such as, “You’re making good progress. You’ll be as good as new,” left a lasting impression on 1 participant.

Even broad statements, like the reminder to “be as healthy as you can for surgery,” had a profound influence on some participants. One participant shared that after hearing this advice, they “immediately came home from that appointment, changed [their] diet, got on the treadmill, and exercised for at least an hour a day, 6 to 7 days a week until surgery.” This commitment led to 13-pound intentional weight loss by the time of cystectomy. The combination of simple prompts and positive feedback from their doctors helped participants build and sustain better exercise habits.

3.2.3. Belief that exercise provides control over their disease course

Participants frequently expressed feelings of “losing a sense of control” during their bladder cancer treatment. Their bodies were behaving differently, and their usual routines were disrupted. Healthcare appointments, missing work, and skipping social events made them feel disconnected from their normal lives. However, exercise became a source of empowerment. As 1 participant shared, “Exercise got me mentally prepared that I can do this surgery. I felt like, I’ve got this. I’m going to get through this. I’m not just going to survive the surgery; I’m going to survive beyond it.” Through exercise, participants regained a sense of control over their bodies and lives. It gave them the confidence to physically endure the challenges of treatment while also strengthening their mental resilience to face their cancer head-on.

3.2.4. Having upcoming healthcare appointments

Approximately one-third of participants reported increased motivation to exercise prior to an upcoming healthcare appointment. This motivation stemmed from 2 main sources. First, they reported wanting to be physically prepared for an upcoming procedure or infusion. As one explained, “I’m motivated when there’s something important coming up. If I have a procedure scheduled, that drives me to get back on the treadmill and work hard to prepare for the surgery.” Second, participants wanted to demonstrate their progress to their doctors. They felt accountable to their healthcare providers and were motivated to show that they had been active. One shared, “You want to be able to tell them that you were successful. I didn’t want to come back and say, ‘Well, I did nothing.’” The combination of personal health goals and the desire to impress their doctor supported participants in this cohort in maintaining their physical activity.

3.4. Attitudes towards digital health care technologies

3.4.1. Attitudes towards wearables (e.g. FitBit®, Apple™ Watch)

Participant experience with wearable physical activity monitors varied widely. While 5 participants (56%) had previously used one to track steps or heart rate, only 3

(33%) had used one consistently. Some found wearables unnecessary, especially if they exercised on machines (e.g. treadmills, ellipticals) that tracked metrics like calories burned and distance travelled. Others found them unhelpful, as they followed familiar outdoor walking routes where distances were known: “I know what my walking courses are, and I know how far they are, and I know what it is without the machine telling me. So why do I need this [wearable] to tell me about it.” Others simply were not interested in knowing their health metrics. However, many participants emphasized the importance of wearables, noting that “the more knowledge you have, the better” and that these devices help maintain consistency with being active.

Despite differing views, all 9 participants agreed that they would use a wearable if it were part of an exercise research study. Those who previously used wearables on a consistent basis were enthusiastic about participating, stating that it would enhance their recovery and “force them to be accountable to exercise.” Even those without prior experience with wearables were open to using them if recommended by their doctor for recovery and well-being.

3.4.2. Attitudes towards an exercise app

App features those participants found most useful (or would like to see) in a mobile exercise app:

1) Doctor provides feedback (both positive and negative) on fitness progress

Nearly all participants expressed that an exercise app would be most valuable if their doctor could provide feedback on their fitness levels and exercise habits. Such feedback would make participants feel “worth the doctor’s time” and offer inspiration throughout their exercise journey. Knowing that their doctor was monitoring their progress would create a sense of accountability and encouragement, with 1 participant stating that check-ins from providers “mean more to me than anything.”

Participants were open to receiving constructive criticism, especially if it helped them correct their form and avoid developing “bad exercise habits.” They actively sought suggestions to improve the effectiveness of their workouts, emphasizing the importance of “getting it right.” However, participants also highlighted the need for positive reinforcement. Simple, encouraging messages like “Great work this week!” or “Keep it up” were seen as highly motivating. One participant noted, “It’s always good to have a little positive encouragement” during recovery.

2) Ability to track fitness progress

Participants expressed a strong desire for the app to track their physical fitness levels, serving as a barometer of their progress. They wanted to know if their workouts were delivering “measurable health benefits.” Being able to see tangible improvements would provide inspiration and motivation to remain active and continue working toward further improvements in their fitness.

3) Competition with other bladder cancer survivors

Many participants expressed a desire to compare their exercise metrics with others who had received similar treatment to themselves. This “friendly competition” would motivate them to stay active by allowing them to “see what other people are doing, and see how you rank.” For others, these comparisons would inspire participants by showing that “exercising is possible” and encouraging the mindset of “if they can do it, why can’t I?” Additionally, this feature could help build a sense of community among participants, creating a network of like-minded individuals working toward a shared goal.

4) Notifications from app are not excessive

When asked about the frequency of notifications from an exercise app, participant responses varied. Some were unbothered by frequent notifications, with one remarking, “In the world we live in, we get notifications 24/7, so it makes no difference.” A few even preferred frequent reminders, stating, “We all need encouragement to get off our rear end.”

However, the majority of participants favored fewer notifications, as frequent messages were seen as “annoying,” with some feeling that they should be motivating themselves without a notification’s prompting. Overall, most participants preferred receiving notifications a few times per week, finding this to be the right balance between gentle reminders and avoiding unnecessary interruptions.

4. Discussion

In this qualitative study, we identified several facilitators and barriers to physical activity among bladder cancer patients who had undergone NAC and RC. We also identified app features those participants with a history of bladder cancer treatment desired, shedding insight into exercise motivations and informational needs for this patient population. Combining these insights revealed that nearly all patients expressed a desire for a structured exercise plan that outlines the duration, intensity and type of exercise to perform. Participants also noted that receiving encouragement and reminders from their healthcare providers would motivate them to stay active. Many also wanted a way to gauge their fitness levels, allowing them to track their progress and compare their fitness to others in similar situations. Collectively, these findings provide new insights into the motivational factors that influence participation in and adherence to physical activity in patients with bladder cancer.

A consistent theme voiced by participants was the need for a prescription outlining a detailed and personalized exercise program. Research shows that these personalized exercise programs improve fitness levels more effectively than general advice such as “take more steps” [13]. To craft these programs, oncology providers may refer to published repositories such as that created by Jensen et al., which contains 22 exercises for fully-mobile cancer patients in the perioperative period [14]. These exercises (6–10

repetitions each, 15–20 minutes total), combined with 30 minutes of daily aerobic activity, have been shown to be feasible and effective in improving functional fitness in patients awaiting cystectomy [15]. In addition, oncologists may also utilize the 8 tips provided in Table 2, which provide actionable items to increase physical activity based off themes collated from this study. However, we acknowledge that oncology providers may not always have the clinic time, knowledge, or capacity to provide detailed exercise prescriptions [16–18]. In these situations, oncologists should consider referrals to cancer rehabilitation specialists such as rehabilitation physicians, physical therapists, and exercise physiologists. These aforementioned providers have dedicated training on how to provide exercise counseling, as well as the expertise in tailoring exercise programs for patients prior to, during, and after bladder cancer treatment. Furthermore, oncologists may consider incorporating exercise during chemotherapy infusion appointments. Prior studies have demonstrated that low-intensity cycling performed during chemotherapy infusions is safe, feasible, and facilitates physical activity during an otherwise sedentary wait time [19,20].

Oncologists must also understand the impact they have on their patient's motivation and adherence to exercise. It is well established that patients are more likely to exercise if recommended by their oncologist [21,22]. Even simple encouragements to exercise can lead to significant increases in patient physical activity [23], and brief encouragements such as “You should be moving around more,” made lasting impressions on patients in our study. Yet, it is not enough to

simply recommend exercise. To optimize physical activity levels in patients, providers should continue to reinforce and remind patients of the importance of staying active.

For example, 1 randomized study found that daily text message reminders increased step counts by 10% to 16% in those undergoing exercise rehabilitation [24]. Even less frequent reminders have proven beneficial – patients who participated in exercise programs with 5 or more provider check-ins exercised significantly more than those with fewer interactions. [23] These “check-ins” create a sense of accountability [25], a statement echoed by participants in our study. Even the mere prospect of an upcoming appointment with a provider can also motivate patients to exercise, a phenomenon known as the “fresh start effect,” where individuals feel more driven to engage in aspirational behaviors around temporal landmarks, such as a doctor's visit [26].

A commonly voiced patient request was for a means to measure their fitness and track progress. This would serve as both a motivator to engage in physical activity more regularly, and conversely, a prompt to increase activity if their fitness was not improving. Providing patients with objective measurements of their fitness has known benefits. Two randomized studies found that surgical patients were more active when given activity trackers and made aware of their step counts, compared to those who were not informed of their step counts [27,28]. Objective assessments of physical capabilities also help patients set realistic fitness goals, which is one of the strongest motivators for behavioral change. [29] Moreover, quantifying physical activity levels may enable patients to compare their fitness with others, a

Table 2
Eight actionable items for providers to increase physical activity in patients.

Actionable items	Example	Associated benefits
Frequently ask patients about their current physical activity levels.	“How much walking are you doing?”	Reinforces that staying active is important, that the provider believes it is beneficial to a patient's health.
Encourage activity and exercise.	“You should be moving around more at home.”	Motivates patients to be active.
Provide a structured workout routine outlining which exercises to perform, with exact duration/repetitions.	“Perform 4 leg lifts per day, 3 dips per day, and walk 1.5 miles 4 times per week.”	Provides concrete exercise plan, removes guesswork, creates actionable goals for patients to achieve.
Give constructive criticism, but prioritize positive reinforcement.	“You are a little behind schedule for where I would expect you to be from a physical fitness standpoint” and/or “Keep it up, great work!”	Inspires patients by letting them know their provider cares and is paying attention to their physical activity levels.
Increased frequency of follow-up appointments if worried about physical activity levels.	“I will see you back in 1 month for a follow up appointment. I want to see that you have been walking 1 mile per day since then.”	Appointments increase accountability to stay on track.
Do not underestimate the power of a doctor's recommendation, especially when framed in a way to improve a patient's health.	“I really want you to adhere to this exercise program, as it will have long-term benefits for your recovery and health.”	Patients view providers recommendations as “prescriptions.”
Track a patient's fitness progress at each visit, and show them how much they are progressing.	“Great job. Three months ago you could barely walk a few blocks. Now I see you are walking a few miles each day.”	Seeing tangible improvements inspires patients to continue exercising. Conversely, not seeing improvements can motivate them to exercise more.
Provide benchmarks of how their fitness levels compare to others in similar situations.	“Other patients of mine who also completed neoadjuvant chemotherapy are walking 2 miles per day in preparation for surgery. You can do this too.”	Fosters competition, inspires patients that exercising is possible.

factor participants believed would motivate them to exercise more. This idea is supported by studies showing that social comparisons of physical activity levels can significantly increase exercise participation and encourage healthy exercise habits [30].

5. Conclusion

Our findings provide insight into the motivations of patients who have undergone NAC/RC for bladder cancer with regards to staying physically active during and after treatment. Providers must be aware of patients' preferences, as well as their own lifestyle and exercise habits, as those who exercise more are more likely to recommend exercise to their patients [31]. Moving forward, providers may utilize the actionable tips provided to effectively encourage activity in those undergoing NAC/RC. Although these tips may seem generalizable to all patients, including those who are not undergoing NAC/RC, it should be noted that these patients represent a unique group with special considerations. Counseling patients undergoing NAC/RC warrants a tailored approach; therefore, these qualitative findings should be used to uniquely optimize fitness and prevent debilitation in this vulnerable group. Ultimately, these findings can be used to guide the development and implementation of exercise programs and digital health technologies for bladder cancer patients.

Declaration of competing interest

The authors declare that they have known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

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Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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